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ORIGINAL RESEARCH

Personalized route recommendation for passengers in urban rail transit based on collaborative filtering algorithm

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Abstract
The rapid advancements in information technology and intelligent systems within urban rail transit (URT) systems have highlighted the need for more personalized route recommendations that consider passengers' travel habits. This study aims to address this issue by investigating passenger travel routes alongside other passengers who share similar travel preferences, utilizing collaborative filtering (CF) techniques. The approach involves analyzing historical card data to assess passenger travel profiles, including actual travel time under crowded conditions. By considering both individual passenger preferences and the preferences of similar passengers, a CF algorithm is employed to offer personalized route recommendations. The Shenzhen metro is used as a case study to illustrate the proposed method. The results demonstrate that the proposed approach surpasses traditional route recommendation methods by providing tailored suggestions that align more closely with passengers' travel preferences. These findings emphasize the value of incorporating passenger travel preferences into route recommendation models, thereby enhancing the accuracy and effectiveness of personalized route recommendations within URT systems.

1 | INTRODUCTION

Urban rail transit (URT) has emerged as a central focus in the development of urban transportation due to its numerous advantages, including its high speed, efficiency, and capacity. As URT networks continue to improve, URT systems will feature more intricate network structures and larger scales, accompanied by a substantial surge in passenger demand. As the URT system expands and the areas surrounding URT lines undergo development, an increasing number of commuters will choose URT for their daily travel needs. Consequently, the URT system will witness a significant influx of passengers, prompting a growing concern about effectively managing and guiding passengers within the URT network. The need to address these demands has become increasingly urgent.

As URT networks expand and become increasingly complex, passengers will face a greater number of route options for their journeys. In situations where the URT network structure is relatively simple, there may be limited viable routes based on factors such as shortest travel time or shortest distance. This means that for each origin-destination (OD) pair, passengers' route choices tend to be more uniform, and the impact of passenger guidance systems may not be significant. However, as the URT network grows in size, passengers' route choices become more diverse rather than singular. For the same origin and destination, different routes offer varying characteristics, such as shorter theoretical travel time, fewer transfers, less congestion, and so on. Consequently, passengers with different travel preferences will opt for different routes. In this scenario, without an appropriate passenger guidance system to direct passengers' route choices, there can be an excessive concentration of passengers on the shortest or fastest routes between destinations. This can lead to issues such as overcrowding, transfer congestion, and other problems. Addressing this challenge is crucial for improving the overall passenger experience and ensuring the efficient operation of URT systems. Therefore, in URT systems, providing convenient, fast, and comfortable travel services based on route recommendation for passengers has become a challenging task for operating authorities.

The current methods for route choice and route recommendation in URT systems vary. The simplest route choice model is the shortest path algorithm like Dijkstra's algorithm [1], while popular route choice methods include Logit models such as nested Logit model and multinomial Logit model [2–4]. However, traditional route choice models do not consider passenger preferences and implementation can be challenging. Route recommendation approaches can be broadly categorized into two types: those based on network flows and route characteristics, and those driven by passenger behaviour and preferences. The former focuses on establishing generalized travel cost models [5, 6] for route recommendations, while the latter involves using questionnaires to analyze passenger route preferences [7]. However, the former faces difficulties in calculating actual passenger flows and searching routes based on individual preferences, while the latter faces challenges in collecting private information through surveys. As a result, the development of an intelligent and personalized route recommendation system has become a research focus in URT systems. Collaborative filtering (CF) is a widely adopted recommendation technique in commercial applications [8, 9]. The main concept of CF is to create user profiles, calculate similarity between users, identify top-N similar users as connections, and make recommendations based on the preferences of connected users. Hence, CF methods can be applied to the problem of route recommendation in URT systems. Therefore, in the proposed problem of route recommendation, a passenger's travel profile can be constructed, and the similarity between passengers can be calculated. This information can then be used in CF to recommend suitable routes to passengers.

This paper introduces a method that utilizes historical passenger travel data and employs a CF recommendation algorithm to describe passenger travel profiles and cater to individual route preferences. It is designed to address the challenges faced by cities with significant congestion during peak hours and complex URT networks. By analyzing data sources like Automatic Fare Collection (AFC) data, station data, and route data, URT operators can divert passenger flows and collaborate with map navigation software to offer personalized route recommendations. This method aims to enhance the accuracy and personalization of route suggestions, ultimately improving the overall passenger experience and efficiency of URT systems. Our contributions are as follows:

1) A CF algorithm is used to recommend routes for target passengers by taking into account the travel preference of the target passenger, the travel preference of similar passengers, and the average travel preferences of the OD category.
2) With route preferences, a passenger travel profile is constructed and the cosine similarity is applied to identify similar passengers.
3) The actual travel time for routes under congested conditions is modelled with the calculation of the number of 'left-behind' at stations, which is useful to construct passenger travel profile.

The rest of paper is organized in five sections. Section 2 provides a brief review of recent studies and introduces the research framework of the paper. Section 3 presents a detailed explanation of the passenger travel profile. In Section 4, the route recommendation model is established using CF techniques. Section 5 presents the main results and discussions based on a case study. Finally, Section 6 concludes the paper, summarizing the findings and contributions made in the study.

2 | LITERATURE REVIEW

This section briefly presents an overview of related research in the fields of travel route choice models, route recommendation systems, and CF.

1) Travel Route Choice Models:

   Travel route choice models have been widely employed in traffic assignment to determine the route preferences of commuters within road networks. One of the simplest models is Dijkstra algorithm, while Logit models have gained popularity as the most commonly used approach. As science and technology have progressed, additional algorithms like A* algorithm have emerged for route selection [10]. Moreover, machine learning techniques, such as dynamic route choice, have been utilized to allow passengers to adapt their route preferences based on real-time conditions [11]. Some researchers have also incorporated actual congestion levels into travel time analysis when making route choices [12]. In addition to considering the network, experts and scholars have begun to analyze passengers' behavioural preferences. Traditional methods for predicting passengers' behavioural preferences in travel route choice involve Stated Preference (SP) surveys and Revealed Preference (RP) surveys. However, advancements in sensing techniques and telecommunication technologies have led to the proposal of new methods for collecting travel information from sources like smart card data, mobile phone data, WIFI data, and GPS data [13–15]. Furthermore, there are researchers who employ reinforcement learning and other techniques to analyze passenger preferences in travel route choice [16].

2) Travel Route Recommendation:

   Travel route recommendation can be broadly categorized into two main approaches: those based on network flows and route characteristics, and those driven by passenger behaviour and preferences. In the category of network flows and route characteristics, various studies have been conducted. Some researchers have developed generalized travel cost models to estimate and predict travel times and costs [5, 6]. Integration programming has been proposed in other studies to optimize route recommendations by considering multiple data and factors simultaneously [17, 18]. The modified Logit algorithm has also been utilized to distribute passenger flow and determine optimal routes [19].

In the approach of passenger behaviour and preferences, researchers have focused on modelling factors such as passengers' tolerance coefficients or preferences to provide route recommendations [20–23]. Due to the availability of different data sources, experts and scholars have explored making route recommendations based on specific data types. For instance, some researchers use smart card data to make route recommendations for passengers [24, 25], while others leverage passengers' online data (social relationships) to establish personalized route choice models [26], AFC data has been analyzed to understand URT passengers' route choice behaviour and offer recommendations [27]. Additionally, machine learning and other methods have been employed by some experts for route recommendation [28].

3) Collaborative filtering:

CF is considered one of the most successful techniques in recommendation systems. It utilizes the preferences of similar individuals with shared experiences to recommend information that would interest users. CF was initially used to calculate the similarity between two users' preferences for products and suggest relevant items to users [8, 9]. The algorithm's core function is to analyze a user's behavioural preferences based on their historical data. By determining the similarity among different users, the algorithm identifies the top-N users who exhibit the highest similarity to the target user and recommends items based on their preferences [29]. This approach has been widely applied in various successful recommender systems [30]. However, as the amount of accumulated data increases over time, two significant challenges arise in user data: data sparsity and the cold-start problem. Data sparsity refers to situations where users do not continuously generate data and produce data across different categories, resulting in low data frequency over a broad timeline [31]. The cold-start problem occurs when a user encounters a new item for which no historical information is available, making it challenging to provide recommendations [32]. Furthermore, most CF methods primarily focus on product and user similarity and often overlook the personalized needs of individual users. They emphasize finding similarities between products and users without fully considering the unique preferences and requirements of each user.

In summary, the aforementioned literature has contributed extensively to the research on travel route choice and recommendation. However, existing research on URT route recommendation lacks consideration for three critical issues

1) The existing recommendation systems seldom consider passengers' travel habits and fail to provide targeted route guidance based on individual passengers' travel preferences.
2) Previous studies on route selection and recommendation mainly rely on theoretical models such as Logit models, which consider the theoretical travel time of routes but do not account for the actual travel time delays caused by congestion.
3) CF algorithms, despite being successful in recommendation systems, have not been applied in the field of URT.

Based on the analysis mentioned above, this paper presents a novel approach to describe passenger travel profiles using historical passenger travel data and POI (point of interest) information. The primary objective is to develop a personalized route recommendation system specifically designed for URT passengers. To achieve this goal, the paper proposes a CF recommendation algorithm that takes into account both the target passenger's route preferences and the preferences of similar passengers. By considering these factors, the algorithm aims to provide more accurate and personalized route recommendations to passengers. This is expected to enhance the overall passenger experience and improve the efficiency of URT systems. The main framework of the study is depicted in Figure 1, which outlines the key components and steps involved in the proposed approach.

3 | PASSENGER TRAVEL PROFILE

In the transportation field, the concept of passenger travel profile is similar to the user profile used in the field of internet browsing. It involves analyzing passengers' travel information to determine their travel preferences. The passenger travel profile typically consists of both traffic characteristics and passenger characteristics. Traffic characteristics encompass various factors such as station categories, route information, congestion levels, and more. Passenger characteristics, on the other hand, include personal basic information (such as age, gender etc.), economic conditions, and specific travel preferences. This study focused on URT systems, the passenger travel profile is composed of four main components: origin station category (PO), destination station category (PD), inbound period (PT), and route preference (PP). Origin station category (PO) indicates the category of the station from which the passenger entered the URT system. Destination station category (PD) represents the category of the station where the passenger exits the URT system. Inbound period (PT) refers to the time period during which the passenger travels between the origin and destination stations. Route preference (PP) signifies the passenger's preference for selecting a specific route within the URT system. The passenger travel profile is typically represented as a vector, as shown in Equation (1).

$$P = \begin{pmatrix} PO & PD & PT & PP \end{pmatrix} \tag{1}$$

For modelling convenience, Table 1 lists all the relevant notations and parameters used in the formulation.

3.1 | Origin/destination station category PO, PD

The land use structure of a city plays a significant role in shaping the characteristics of URT stations. Each station is unique but also shares similarities with others based on their surrounding land use. Understanding these similarities can provide valuable insights into the patterns of passenger travel between different stations. For instance, if a station is located in a Residential area,

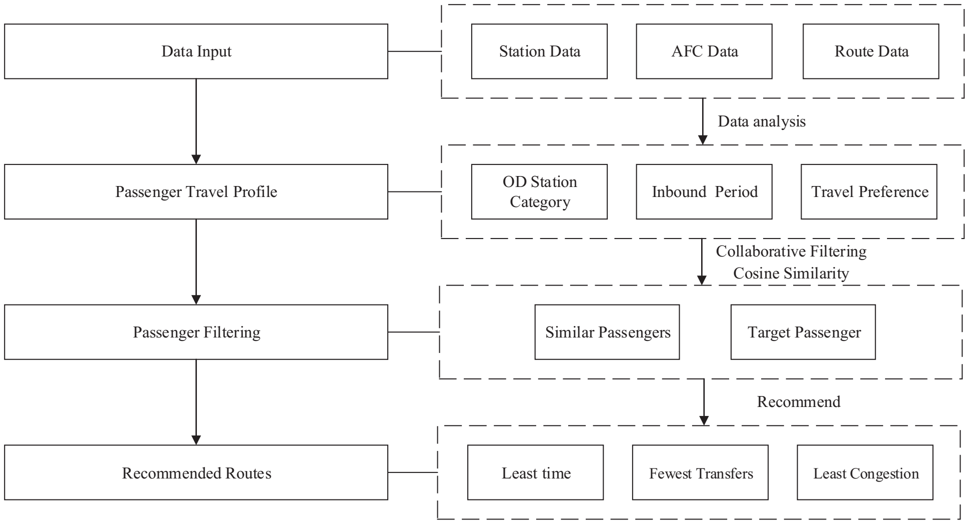

FIGURE 1 The main framework of the study.

TABLE 1 Notations and parameters used in the formulation.

Notation	Definition
ab	Station index
i	Travel route index
j	The order of the origin or transfer stations in route i, where $j = 0$ denotes the origin station of the route, and $j > 0$ denotes the transfer stations of the route
k	The order of the station j in the station list of the URT line
$R_{ab,i}$	The route feature vector from station a to b using route i
$T_{ab,i}^{\text{actual}}$	Actual travel time of the passenger from stations a and b using route i;
$T_{ab,i}^{\text{travel}}$	Theoretical travel time from stations a to b using route i
$T_{ab,i}^{\text{trans}}$	Total transfer walking time from stations a to b using route i
$T_{ab,i}^{\text{delay}}$	Additional delay time caused by 'left-behind' from stations a to b using route i
$N_{ab,i}^{\text{trans}}$	Total transfer times required for passengers to travel from stations a to b using route i
p_{ab}^{time}	The probability that a passenger chooses a route with the least time from stations a to b
p_{ab}^{trans}	The probability that a passenger chooses a route with the fewest transfers from stations a to b
p_{ab}^{crowd}	The probability that a passenger chooses a route with the least congestion from stations a to b
$L_{ab,i}$	The total number of 'left behind' due to congestion from stations a to b using route i
$\overline{l_{ab,i,j}}$	The number of theoretical 'left-behind' due to congestion for the jth transfer from stations a to b using route i in the study period
$C_{ab,i,j}$	Train capacity of the URT line for the jth transfer from stations a to b using route i
$P_{ab,i,j,k}^{\text{bo}}$	The number of passengers boarding trains who enter station k for the jth transfer from stations a to b using route i
$P_{ab,i,j,k}^{\text{trans-bo}}$	The number of passengers boarding trains who transfer from other URT lines at station k for the jth transfer from stations a to b using route i
$P_{ab,i,j,k}^{\text{al}}$	The number of passengers alighting trains who exit station k for the jth transfer from stations a to b using route i
$P_{ab,i,j,k}^{\text{trans-al}}$	The number of passengers alighting trains who transfer to other URT lines at station k for the jth transfer from stations a to b using route i
$l_{ab,i,j}$	The number of 'left-behind' used in the calculation for the jth transfer from stations a to b using route i
$l_{\max}$	The maximum number of 'left-behind', which can be determined based on the actual situation of the URT system
$H_{ab,i,j}$	Train headway in the URT line for the jth transfer from stations a to b using route i
$PP_{ab,1}$	The target passenger's route preferences
$R_{a'b'}^{\text{time}}$ $R_{a'b'}^{\text{trans}}$ $R_{a'b'}^{\text{crowd}}$	The least time route, the fewest transfer route, and the least congestion route of the OD category $a'b'$, respectively

TABLE 2 Category classification of Shenzhen metro stations.

Station categories	Station name
Residential	Shixia, Gushu, Xixiang, Xiangxi Village…
Corporate Enterprises	Huaqiang North, Gangxia North, Gaoxin Park…
Food and Beverage	Old Street, Chegongmiao, Shopping Park…
Commercial Retail	Bamboo Grove, Huaqiang North, Old Street, Yuanling…
Service Industry	Buji, Pingshanwei, Guomao, Mu mianwan…
Education	Liuxiandong, Shenda, Xili, Longhua…
Medical Services	Jingtian, Lianhua West, Chi Mei, Convention and Exhibition Center…
Recreation	Shopping Park, Civic Center, Science Museum…
Tourist Attractions	Xiangmi Lake, Dongzhong Memorial Hall, Xianhu Road, Liantang…
Government	Civic Center, Guomao, Renmin South, Gangxia…
Transportation	Futian, Airport East, Shenzhen North, Buji…
Sports Center	Danyun, Danyun Center…

it implies that there are more residential neighbourhoods nearby. As a result, the station is likely to experience higher inbound traffic in the morning as residents commute to work or school, and higher outbound traffic in the evening as they return home. Conversely, a station categorized as Commercial Retail may have more shopping malls in its vicinity, leading to increased traffic on weekends compared to weekdays.

To achieve a more detailed classification of URT stations, this paper utilizes POI classification data obtained from a public map developer platform. The POI data is organized into three levels of classification, including 23 first-level categories, 267 second-level categories, and 869 third-level categories. In this study, 23 first-level categories will be used as POI indicator categories. By leveraging these POI categories, we can analyze and compare the characteristics of different URT stations. This analysis provides insights into the behaviour and travel patterns of passengers commuting between these stations. By considering the specific land use characteristics and associated POIs, it becomes possible to develop a more comprehensive understanding of passenger travel within the URT system.

By utilizing the POI indicators and the passenger flow of URT stations at different times of the day, the K-means algorithm is applied to cluster the URT stations to the above categories. This clustering results in 12 categories for URT stations, such as Residential, Corporate Enterprises, Food & Beverage, Commercial Retail, Service Industry, Education, Medical Services, Recreation, Tourist Attractions, Government, Transportation, and Sports Center. Table 2 presents the category classification of Shenzhen metro stations based on the clustering results. These categories serve as the basis for generating passenger travel profiles that accurately reflect their preferences.

3.2 | Inbound period *PT*

The inbound period in this study refers to the period when passengers travel between their origin and destination within

TABLE 3 Categories of inbound period.

	Peak period		
Time	Morning peak	Evening peak	Off-peak period
Weekday	7:30–9:30	16:30–19:00	10:00–12:00 20:00–21:00
Weekend	9:00–19:00		7:30–9:00 19:00–21:00
Holiday	9:00–19:00		7:30–9:00 19:00–21:00

the URT system. The paper categorizes the inbound period into seven distinct categories: weekday morning peak, weekday evening peak, weekday off-peak, weekend peak, weekend off-peak, holiday peak, and holiday off-peak. When a holiday coincides with a weekday or weekend, the holiday period takes priority. The specific time periods for each category can be found in Table 3.

3.3 | Routing preferences *PP*

Passenger route preference refers to the probability of passengers selecting different travel routes based on their individual preferences. When travelling within the URT system, passengers often have multiple route options available to them, each with its own characteristics, such as travel time, transfer times, congestion levels etc. To represent these various features associated with different routes, a route feature vector denoted as $R_{ab,i}$ is introduced to represent the passenger's travel from station a to b using route i is introduced, as shown in Equation (2):

$$R_{ab,i} = \left(T_{ab,i}^{\text{travel}}, N_{ab,i}^{\text{trans}}, L_{ab,i}\right) \tag{2}$$

The feature vector can be categorized into three types, as indicated in Equation (3). These categories are: (1) the least time route, denoted as R_{ab}^{time}, represents the route that minimizes travel time for passengers; (2) the fewest transfers route, denoted as R_{ab}^{trans}, represents the route that requires the least number of transfers for passengers; (3) the least congestion route, denoted as R_{ab}^{crowd}, represents the route that minimizes congestion for passengers. It is important to note that these three categories can overlap, meaning that certain routes can satisfy two or even all three types simultaneously. For instance, there may be routes that are both the fastest and least congested, or routes that require the fewest transfers while also having low congestion.

$$\begin{cases} R_{ab}^{\text{time}} = R_{ab,i} & i = \arg\min_i T_{ab,i}^{\text{travel}} \\ R_{ab}^{\text{trans}} = R_{ab,i} & i = \arg\min_i N_{ab,i}^{\text{trans}} \\ R_{ab}^{\text{crowd}} = R_{ab,i} & i = \arg\min_i L_{ab,i} \end{cases} \tag{3}$$

Once the route feature vector has been established, it becomes possible to construct a route preference matrix, denoted as PP_{ab}, which is based on the passenger's historical travel data, as shown in Equation (4).

$$PP_{ab} = \begin{pmatrix} R_{ab}^{\text{time}} & R_{ab}^{\text{trans}} & R_{ab}^{\text{crowd}} \\ p_{ab}^{\text{time}} & p_{ab}^{\text{trans}} & p_{ab}^{\text{crowd}} \end{pmatrix} \tag{4}$$

To calculate the probability of passenger route selection, the *K*-means clustering algorithm is utilized in the following steps:

1) Categorization of Travel Routes: The travel routes from stations a to b are categorized into three types based on their characteristics: the least time, the fewest transfers, and the least congestion. The actual travel time $T_{ab,i}^{\text{real}}$ for each route is calculated, which will be discussed further in Section 3.4 of the paper.
2) Clustering with *K*-Means Algorithm: The actual travel time for each route is used as the clustering centre, and the historical travel data of the target passenger from stations a to b is utilized as the clustering data source. The *K*-means algorithm is then applied to cluster the data, assigning each trip to the nearest cluster centre.
3) Calculation of Route Preference Matrix: The number of trips for each category of route is divided by the total number of trips made by the passenger. This division process yields the probability of each route category for the passenger. Consequently, the passenger route preference matrix is generated.

3.4 | Actual travel time of routes under congestion

Currently, theoretical travel time is commonly used to estimate passengers' travel time. However, it is crucial to consider that the actual travel time of different routes can vary due to congestion, which subsequently impacts passengers' route preferences. Congestion can occur when the URT network and stations become overcrowded, leading to situations where passengers are unable to board trains, which is called 'left-behind'. When congestion happens, passengers may be unable to board the current train due to overcrowding, which requires them to wait for the next train. In cases of severe congestion, passengers may even have to wait for multiple trains, resulting in a longer actual travel time on their routes. Therefore, it becomes crucial to determine the actual travel time of routes under congested conditions in order to provide accurate and reliable passenger route recommendations. By incorporating the actual travel time, which reflects the impact of congestion, it can offer more realistic and effective route suggestions tailored to each passenger's preference. Calculating the actual travel time under congested conditions is the initial step towards achieving this goal. Equation (5) is used to describe the actual travel time of a route under congestion.

$$T_{ab,i}^{\text{actual}} = T_{ab,i}^{\text{travel}} + T_{ab,i}^{\text{trans}} + T_{ab,i}^{\text{delay}} \tag{5}$$

3.4.1 | The calculation for the number of 'Left-Behind'

A detailed explanation of the calculation method for the number of 'left-behind' is presented in this section. Assuming that passenger arrivals follow a uniform distribution. The number of theoretical 'left-behind' can be expressed using Equation (6).

$$\overline{l_{ab,i,j}} = \frac{P_{ab,i,j,k}^{\text{bo}} + P_{ab,i,j,k}^{\text{trans-bo}}}{C_{ab,i,j} - \sum_{s=1}^{k-1}\left(P_{ab,i,j,s}^{\text{bo}} + P_{ab,i,j,s}^{\text{trans-bo}} - P_{ab,i,j,s}^{\text{al}} - P_{ab,i,j,s}^{\text{trans-al}}\right) + P_{ab,i,j,k}^{\text{al}} + P_{ab,i,j,k}^{\text{trans-al}}} - 1 \tag{6}$$

Using Figure 2 as an example, we describe the meanings of various relevant parameter symbols and illustrate the passenger's travel trajectory on the URT network. In Figure 2, a passenger who enters the station at station a and takes two transfers along the red trajectory line, passing through three different lines before finally exiting at station b. At station a, the passenger enters the station and boards the train. At this station, the parameter of station index is $j = 0, k = k_1$. The passenger waits for the train along with the boarding passenger flow $P_{ab,i,0,k_1}^{\text{bo}}$ and calculates the number of 'left-behind' before boarding. Upon reaching the first transfer station, the passenger alights and transfers to Line II. At this transfer station, the parameter of station index is $j = 1, k = k_2$. Taking into account both the boarding passenger flow $P_{ab,i,1,k_2}^{\text{bo}}$ and transfer-in passenger flow $P_{ab,i,1,k_2}^{\text{trans-bo}}$ at this station, as well as the alighting passenger flow $P_{ab,i,1,k_2}^{\text{al}}$ and transfer-out passenger flow $P_{ab,i,1,k_2}^{\text{trans-al}}$ from the train, the number of 'left-behind' is calculated before the passenger boards the next train. When arriving at the second transfer station, the passenger alights and transfers to Line III. At this transfer station, the parameter of station index is $j = 2, k = k_3$. Similarly, considering the boarding passenger flow $P_{ab,i,2,k_3}^{\text{bo}}$ and transfer-in passenger flow $P_{ab,i,2,k_3}^{\text{trans-bo}}$ at this station, along with the alighting passenger flow $P_{ab,i,2,k_3}^{\text{al}}$ and transfer-out passenger flow $P_{ab,i,2,k_3}^{\text{trans-al}}$ from the train, the number of 'left-behind' is calculated before the passenger boards the next train. Finally, the passenger reaches the destination station b. This description outlines the passenger's travel trajectory and the calculation of 'left-behind' at each stage, considering various factors such as passenger flows, transfers, and train operations.

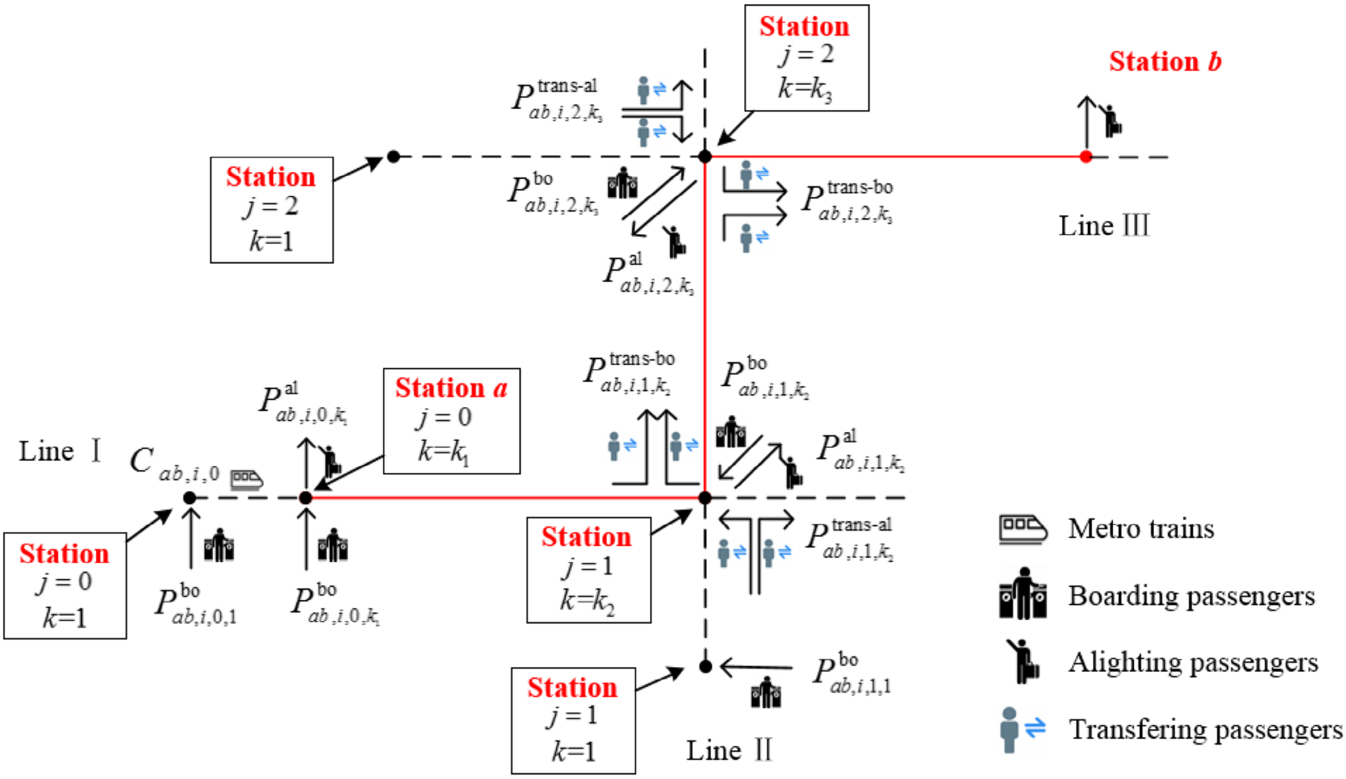

FIGURE 2 The example of passengers' travel trajectory between the OD pair.

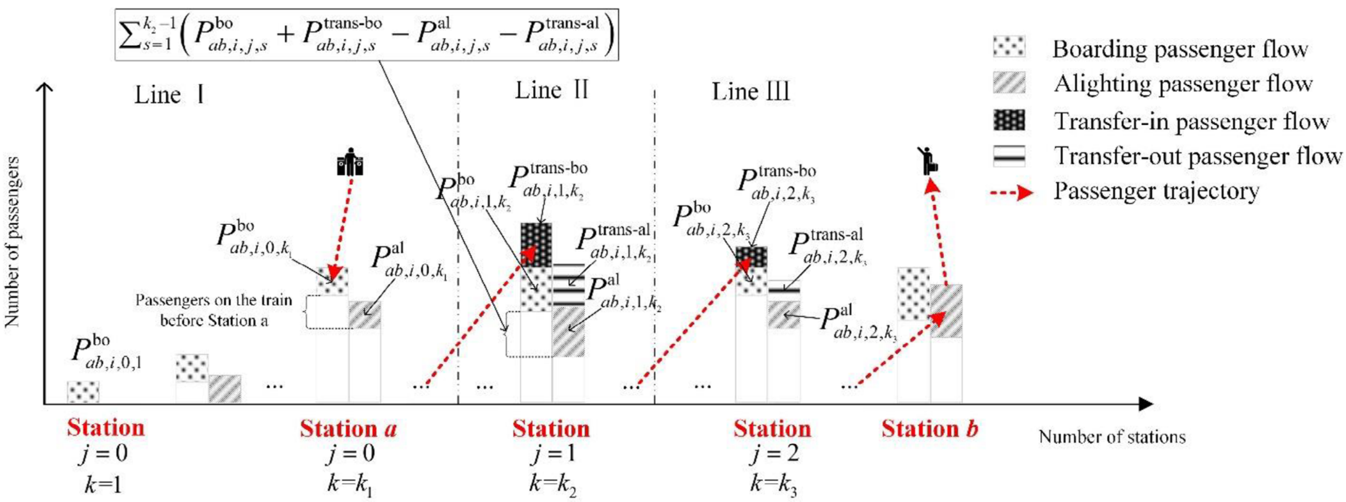

FIGURE 3 The composition of arriving train flows and the calculation of 'left-behind'.

Based on the passenger's travel trajectory example in Figure 2, Figure 3 is used as an example to provide a detailed explanation of the passenger flow composition of arriving trains and the calculation of 'left-behind' at each station. The passenger starts at station a, takes two transfers, and exits at station b. The passenger flow composition at the origin station and transfer stations can be observed from the bar chart in Figure 3. In the figure, each bar with two columns represents a station. The left side represents boarding and transfer-in passenger flow, while the right side represents alighting and transfer-out passenger flow. The white area at the bottom of the two columns represents the cumulative boarding and alighting passenger flow before the train arrives at that station. The coloured and shaped rectangles represent the boarding, alighting, and transferring passenger flow at the stations. The difference between the two represents the passenger flow on the train upon arriving at that station. For example, at station $j = 1, k = k_2$ on Line II, as shown in Figure 2, the passenger flow on the train that arrived at that station during the study period is

$$\sum_{s=1}^{k_2-1}\left(P_{ab,i,j,s}^{\text{bo}}+P_{ab,i,j,s}^{\text{trans}-\text{bo}}-P_{ab,i,j,s}^{\text{al}}-P_{ab,i,j,s}^{\text{trans}-\text{al}}\right)$$

When the growth of the first column is greater than the second column in the chart, it indicates an increase in the number of passengers on the train when passing through that station. Conversely, when the growth of the second column is greater than the first column, it indicates a decrease in the number of passengers on the train when passing through that station. Therefore, when the number of passengers on the train reaches its capacity, subsequent passengers are unable to board, and they have to wait for the next available train. Based on the composition of arriving train flows shown in Figure 3, the theoretical calculation for the number of 'left-behind' can be obtained using Equation (6).

3.4.2 | The calculation of route delay time and total number of 'left-behind'

To account for the theoretical range of possible values for the number of 'left-behind', Equation (7) is introduced to limit them based on the actual situation of the URT system. For the route from stations a to b using route i, the delay time experienced by passengers can be represented by Equation (8), and the total number of 'left-behind' for passengers can be represented by Equation (9).

$$l_{ab,i,j}=\begin{cases} l_{\max} & \overline{l_{ab,i,j}}\le -1 \\ 0 & -1<\overline{l_{ab,i,j}}\le 0 \\ \overline{l_{ab,i,j}} & 0<\overline{l_{ab,i,j}}\le l_{\max} \\ l_{\max} & \overline{l_{ab,i,j}}>l_{\max} \end{cases} \tag{7}$$

$$T_{ab,i}^{\text{delay}}=\sum_j l_{ab,i,j}\times H_{ab,i,j} \tag{8}$$

$$L_{ab,i}=\sum_j l_{ab,i,j} \tag{9}$$

In Equation (7), $\overline{l_{ab,i,j}}\le -1$ and $\overline{l_{ab,i,j}}>l_{\max}$ indicate extreme congestion scenarios on the travel route. $l_{\max}$ represents the maximum number of 'left-behind', which occurs when the level of congestion is at its peak, and the trains have already reached their full capacity before arriving at the station. In such cases, the number of 'left-behind' is set to the maximum value determined by $l_{\max}$. On the other hand, $-1<\overline{l_{ab,i,j}}\le 0$ represents a scenario with lower congestion, where the trains have a lower occupancy rate and provide more available positions for passengers. In this situation, passengers do not need to be 'left-behind' as there is sufficient space on the trains for everyone. Additionally, $0<\overline{l_{ab,i,j}}\le l_{\max}$ represents a more congested scenario where the number of 'left-behind' is not at its maximum, but passengers may still need to wait for several train headways in order to board the coming trains due to the higher demand compared to the available capacity.

4 | PASSENGER ROUTE RECOMMENDATION USING COLLABORATIVE FILTERING

4.1 | Collaborative filtering algorithm

This study introduces the CF algorithm for the research of recommending passenger travel routes in URT systems. The CF algorithm groups together similar items with similar preferences, based on the principle that users are likely to prefer items that are favoured by other users with similar preferences. In the URT, passengers have diverse preferences when it comes to selecting their travel routes, which is analogous to users' preferences for product selection in e-commerce platforms. Therefore, the CF recommendation algorithm is a suitable method for recommending travel routes to passengers. It can analyze the travel data of different passengers, identify similar patterns, and recommend routes that are likely to be preferred by passengers with similar preferences.

4.2 | Route recommendation using collaborative filtering

The framework of route recommendation based on CF is shown in Figure 4.

4.2.1 | Feature vectors of routes

Assuming a target passenger travelling from the origin station a to the destination station b, a travel route is scheduled to recommend for this passenger. The first step is to find the possible list of travel routes between the OD pair and calculate the feature vectors of these routes within the inbound period. By utilizing Equations (2) and (3) from Section 3.3, we can compute and obtain the feature vectors of the travel routes between the OD pair.

4.2.2 | Target passenger route preference

The historical AFC data of the passenger is utilized to cluster and analyze using the K-means clustering algorithm, as described in Section 3.3. This analysis enables us to derive the route preferences of the target passenger within the OD pair, as indicated by Equation (10).

$$PP_{ab,1}=\begin{pmatrix} R_{ab}^{\text{time}} & R_{ab}^{\text{trans}} & R_{ab}^{\text{crowd}} \\ p_{ab,1}^{\text{time}} & p_{ab,1}^{\text{trans}} & p_{ab,1}^{\text{crowd}} \end{pmatrix} \tag{10}$$

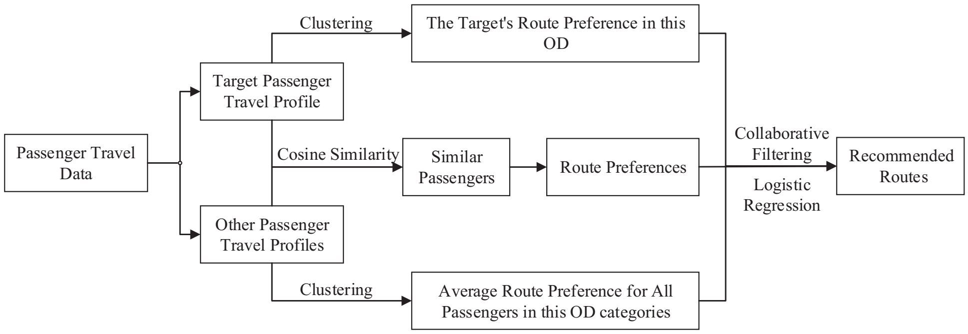

FIGURE 4 The framework of route recommendation based on collaborative filtering.

4.2.3 | Target passenger travel profile

The categories of origin and destination stations are determined by considering the properties of the stations, as explained in Section 3.1. Similarly, the categories of inbound periods are determined based on the time period, as described in Section 3.2. By combining this information with the derived route preferences, we can create the travel profile for the target passenger, which provides a comprehensive understanding of their preferences.

4.2.4 | Similar passengers' route preferences

The determination of similar passengers is primarily accomplished by analyzing the route selection preferences of passengers belonging to the same categories of origin and destination stations, as well as inbound periods. This analysis helps identify passengers who exhibit similar travel preferences to the target passenger. Following the CF approach, the similarity group is obtained by clustering passenger route preferences and calculating the **cosine similarity** between other passengers and the target passenger. The advantage of using cosine similarity in the similarity calculation is that it is unaffected by the number of trips made by passengers. It solely considers the passengers' preferences for selecting routes of different categories, providing a more accurate and realistic measure of similarity.

Assume that the route preference matrices of two passengers are represented by $PP_{ab,1}$ and $PP_{ab,2}$, respectively, where $PP_{ab,1}$ represents the route preferences of the target passenger. The similarity $sim_{ab,1-2}$between them can be calculated using cosine similarity, as shown in Equation (11).

$$sim_{ab,1-2} = \frac{p_{ab,1}^{\text{time}} \cdot p_{ab,2}^{\text{time}} + p_{ab,1}^{\text{trans}} \cdot p_{ab,2}^{\text{trans}} + p_{ab,1}^{\text{crowd}} \cdot p_{ab,2}^{\text{crowd}}}{\sqrt{\left(p_{ab,1}^{\text{time}}\right)^2 + \left(p_{ab,1}^{\text{trans}}\right)^2 + \left(p_{ab,1}^{\text{crowd}}\right)^2} \cdot \sqrt{\left(p_{ab,2}^{\text{time}}\right)^2 + \left(p_{ab,2}^{\text{trans}}\right)^2 + \left(p_{ab,2}^{\text{crowd}}\right)^2}} \tag{11}$$

In order to obtain similar passengers for personalized route recommendations, the TOP-5 passengers with the highest cosine similarity are selected. The travel route preference matrix PP_{ab}^{sim} for the target passenger can be obtained using the following Equations (12) and (13):

$$PP_{ab}^{\text{sim}} = \begin{pmatrix} R_{ab}^{\text{time}} & R_{ab}^{\text{trans}} & R_{ab}^{\text{crowd}} \\ p_{ab}^{\text{time,sim}} & p_{ab}^{\text{trans,sim}} & p_{ab}^{\text{crowd,sim}} \end{pmatrix} \tag{12}$$

$$p_{ab}^{\text{time,sim}} = \frac{\sum_{i=1}^{5} p_{ab,i}^{\text{time,sim}}}{5}, \quad p_{ab}^{\text{trans,sim}} = \frac{\sum_{i=1}^{5} p_{ab,i}^{\text{trans,sim}}}{5},$$
$$p_{ab}^{\text{crowd,sim}} = \frac{\sum_{i=1}^{5} p_{ab,i}^{\text{crowd,sim}}}{5} \tag{13}$$

where$p_{ab,i}^{\text{time,sim}}, p_{ab,i}^{\text{trans,sim}}, p_{ab,i}^{\text{crowd,sim}}, \forall i \in [1,5]$ represent the probability of route selection for the TOP-5 passengers who have the highest similarity to the target passenger in the OD pair.

4.2.5 | Route preferences of all passengers in this category of the OD pair

By filtering the historical travel data of all passengers who have travelled on the specific OD category $(a'b')$, the average route preference matrix $PP_{a'b'}^{\text{avg}}$ can be obtained. Equation (14) is used to represent the average route preference. This step is aimed at providing cold-start route recommendations for passengers who lack previous travel data within the same OD pair. In other words, it refers to passengers for whom historical travel data is unavailable to determine their travel route preferences within the given OD pair. When dealing with cold-start route recommendations for passengers, it is possible to utilize the travel preference of passengers belonging to the same OD category to make route recommendations for the target passenger.

$$PP_{d'b'}^{\text{avg}} = \begin{pmatrix} R_{d'b'}^{\text{time}} & R_{d'b'}^{\text{trans}} & R_{d'b'}^{\text{crowd}} \\ p_{d'b'}^{\text{time,avg}} & p_{d'b'}^{\text{trans,avg}} & p_{d'b'}^{\text{crowd,avg}} \end{pmatrix} \tag{14}$$

where $p_{d'b'}^{\text{time,avg}}$, $p_{d'b'}^{\text{trans,avg}}$, and $p_{d'b'}^{\text{crowd,avg}}$ denote the average passenger's route selection probability for the least time route, the fewest transfer route, and the least congestion route within the OD category ($d'b'$), respectively.

4.2.6 | Personalized route recommendation for the target passenger

To offer travel route recommendations to the target passenger, Equations (15) are utilized, which consider the target passenger's travel route preference, similar passenger route preference, and the route preferences of all passengers in the OD category. Due to the varying volumes of travel data for these three routes, a logistic regression model is employed to generate the route recommendation probability for each route. This probability is used to transform the output values of the logistic regression model into probabilities ranging from 0 to 1. The route with the highest probability is then selected as the recommended route for the target passenger.

$$\begin{cases} p_{ab}^{\text{time}} = \dfrac{1}{1 + e^{\frac{-\left(p_{ab,1}^{\text{time}} + p_{ab}^{\text{time,sim}} + p_{d'b'}^{\text{time,avg}}\right)}{3}}} \\ p_{ab}^{\text{trans}} = \dfrac{1}{1 + e^{\frac{-\left(p_{ab,1}^{\text{trans}} + p_{ab}^{\text{trans,sim}} + p_{d'b'}^{\text{trans,avg}}\right)}{3}}} \\ p_{ab}^{\text{crowd}} = \dfrac{1}{1 + e^{\frac{-\left(p_{ab,1}^{\text{crowd}} + p_{ab}^{\text{crowd,sim}} + p_{d'b'}^{\text{crowd,avg}}\right)}{3}}} \end{cases} \tag{15}$$

where p_{ab}^{time} denotes the recommended value for the route with the least time;

p_{ab}^{trans} denotes the recommended value for the route with the fewest transfers;

p_{ab}^{crowd} denotes the recommended value for the route with the least congestion;

Based on the provided steps, Algorithm 1 can be summarized as follows:

Step 1: Characterize the travel profile of the target passenger

Step 1.1: Generate feature vectors for routes between the OD pair for the target passenger;

Step 1.2: Calculate the route preference of the target passenger using historical AFC data;

Step 1.3: Formulate the travel profile of the target passenger;

Step 2: Identify similar passengers

Step 2.1: Select passengers with similar travel preferences based on cosine similarity using their travel profiles within the OD pair;

Step 2.2: Search the route preferences of all passengers within the same category of the OD pair based on the station types;

Step 3: Route recommendation

Step 3.1: Utilize logistic regression algorithms to determine recommended values for different routes;

Step 3.2: Recommend routes to the target passenger based on the aforementioned values.

5 | CASE STUDY

5.1 | Case data

The case study selects the Shenzhen Metro as an illustration of the proposed algorithm. The dataset used in this case study is comprised of the AFC data for the Shenzhen Metro in April 2021. The dataset comprises approximately 12 million trip records, consisting of AFC smart card data. For analysis purposes, the travel data for a specific passenger with the ID '690409606' was selected as the target passenger. This passenger accumulated a total of 58 trips during the month, as indicated in Table 4. The dataset includes information such as the passenger ID, the origin and destination stations for each trip, and the corresponding timestamps.

5.2 | Algorithm process and case result

By analyzing the AFC data of the target passenger, the commuting OD pair for the route between Houhai and Huaqiangbei was identified. This OD pair consisted of a total of 44 trip data entries. In the Shenzhen Metro network, it was observed that there are multiple travel routes available for this particular OD pair, as depicted in Figure 5. The Houhai station is categorized under Corporate Enterprises, while the Huaqiangbei station also belongs to the category of Corporate Enterprises. Additional information regarding the Houhai—Huaqiangbei travel route, including the station names, distances, and estimated travel times, can be found in Table 5.

1) Route congestion

Route 1 was selected as an example to demonstrate the calculation of the number of 'left-behind' during peak hours. This route entails boarding at Houhai station and transferring at Gangxiabei station. During peak hours, congestion and delays may occur due to the high passenger volume at both stations. By utilizing Equation (6), it is possible to estimate the congestion level and the number of passengers who may not be able to board trains on time along the route. The calculation results for Route 1 are presented in Table 6. It was determined that during peak hours, passengers using Route 1 may have to wait for approximately 1.2 train headways at Gangxiabei station before

TABLE 4 Samples of AFC data for the target passenger.

ID card	Entry station	Entry time	Departure station	Departure time
690409606	1268032000	2021-04-02 08:28:25	126802400	2021-04-02 09:00:03
690409606	1268032000	2021-04-03 08:48:46	126802400	2021-04-03 09:22:01
690409606	1268032000	2021-04-05 08:53:07	126802400	2021-04-02 09:13:35
…	…	…	…	…
690409606	1268032000	2021-04-28 08:18:45	126802400	2021-04-28 08:39:45

AFC, automatic fare collection.

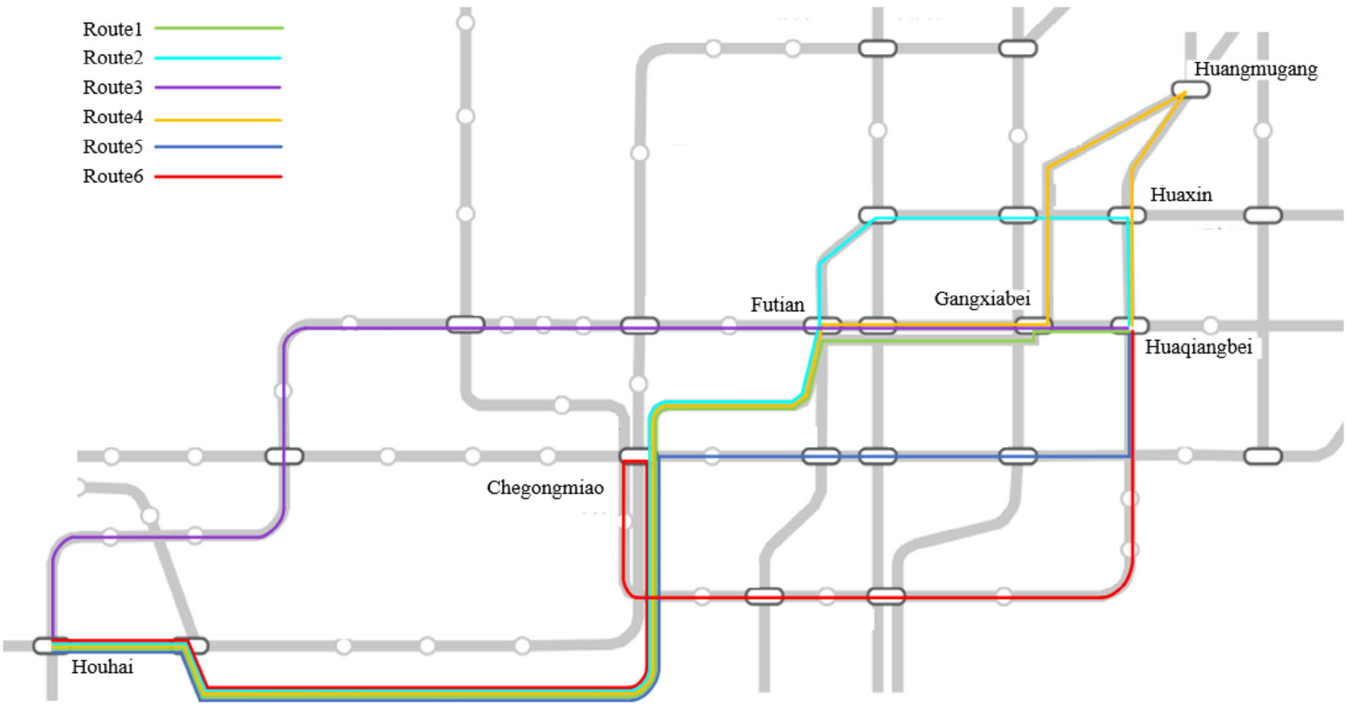

FIGURE 5 Travel routes between Houhai—Huaqiangbei.
Route 1: Houhai—Gangxiabei—Huaqiangbei
Route 2: Houhai—Futian—Huaqiangbei
Route 3: Houhai—Huaqiangbei
Route 4: Houhai—Gangxiabei—Huangmugang—Huaqiangbei
Route 5: Houhai—Chegongmiao—Huaqiangbei
Route 6: Houhai—Chegongmiao—Huaqiangbei

TABLE 5 Routes information of Houhai—Huaqiangbei.

Route no.	Transfer line	Transfer stations	Theoretical time/(s)	Transfer times
1	Line 11–Line 2	Gangxiabei	1440	1
2	Line 11–Line 3	Futian	1920	1
3	Line 2	None	1960	0
4	Line 11–Line 14 –Line 7	Gangxibei, Huangmugang	1800	2
5	Line 11–Line 1	Chegongmiao	1980	1
6	Line 11–Line 7	Chegongmiao	2340	1

TABLE 6 The number of 'left-behind' for Route 1.

Station no.	Station name	Origin station/ Transfer station	Number of 'left-behind' during peak hours
1	Houhai	Origin station	0.6
2	Gangxiabei	Transfer station	1.2

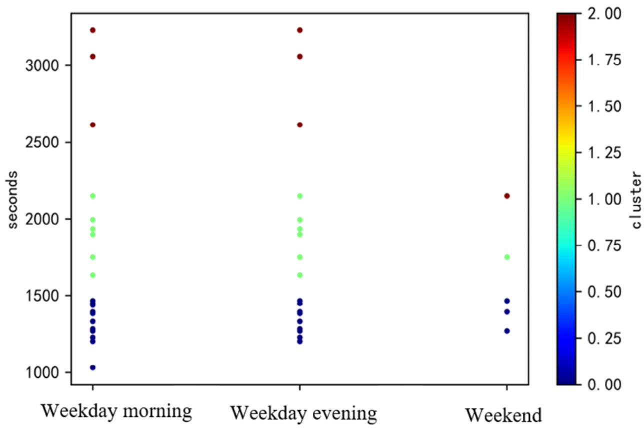

FIGURE 6 Cluster result of passenger route selection in the OD pair.

being able to board a train. In contrast, at Houhai station, the waiting time is estimated to be approximately 0.6 train headways.

Based on Equation (5), the actual travel time for each route can be computed. Furthermore, based on their characteristics, each travel route can be categorized into one of three distinct types. These categories, along with their corresponding characteristics, are outlined in Table 7. It is worth noting that in certain instances, a travel route may satisfy the criteria for all three categories simultaneously, indicating that it possesses attributes from each category.

2) Passenger route preference and travel profiles

The passenger's route preference plays a crucial role in providing personalized travel route recommendations. By analyzing the passenger's historical travel data, specifically the actual travel time for each route, their route preferences can be determined. Regarding the target passenger, the clustering results are depicted in Figure 6. Upon analyzing the passenger's actual travel data, it was discovered that they undertook a total of 44 trips within a month. During weekday mornings, there were 21 trips, and the passenger selected the route with the fewest transfers 12 times, the route with the shortest travel time 6 times, and the route with the least congestion 3 times. Similarly, during weekday evenings, there were 18 trips, and the passenger opted for the route with the fewest transfers nine times, the route with the shortest travel time six times, and the route with the least congestion three times. On weekends, there were five trips, and the passenger chose the route with the fewest transfers three times, the route with the shortest travel time once, and the route with the least congestion once.

After acquiring the passenger's route preferences for the Houhai—Huaqiangbei route, the next step involves generating passenger travel profiles. The outcomes of these calculations are displayed in Table 8.

3) Recommended routes

1. Route preference of the target passenger

According to the data obtained in Table 8, the target passenger route preference $PP_{ab,1}$ in the OD pair is shown as

$$PP_{ab,1} = \begin{pmatrix} R_{ab}^{\mathrm{time}} & R_{ab}^{\mathrm{trans}} & R_{ab}^{\mathrm{crowd}} \\ 0.29 & 0.57 & 0.14 \end{pmatrix}$$

2. Similar passenger route preference

After acquiring the passenger profiles for the OD pair, the proposed algorithm is employed to identify passengers who share similar travel preferences with the target passenger. The cosine similarity is utilized to calculate the similarity between the target passenger and other passengers in the same OD pair. This similarity information is then leveraged to identify the top five users who exhibit the highest similarity to the target passenger. Once the top five similar passengers have been identified, their travel data is analyzed to determine their route preferences PP_{sim}.

$$PP_{ab}^{\mathrm{sim}} = \begin{pmatrix} R_{ab}^{\mathrm{time}} & R_{ab}^{\mathrm{trans}} & R_{ab}^{\mathrm{crowd}} \\ 0.12 & 0.55 & 0.33 \end{pmatrix}$$

3. Average passenger route preference

Collect the route preferences of all passengers who have travelled within the same OD categories $(a'b')$ using a month data.

$$PP_{a'b'}^{\mathrm{avg}} = \begin{pmatrix} R_{a'b'}^{\mathrm{time}} & R_{a'b'}^{\mathrm{trans}} & R_{a'b'}^{\mathrm{crowd}} \\ 0.13 & 0.59 & 0.28 \end{pmatrix}$$

4. Route recommendation

Based on the passenger's route preferences and the analysis of similar passengers, personalized travel route recommendations can be generated. For the target passenger, the route with the fewest transfer is recommended with highest value. In the case at hand, Route 3 (Houhai—Huaqiangbei) emerges as the recommended route, aligning with the passenger's preference for minimizing transfers.

$$\begin{cases} p_{ab}^{\mathrm{time}} = 0.56 \\ p_{ab}^{\mathrm{trans}} = 0.64 \\ p_{ab}^{\mathrm{crowd}} = 0.54 \end{cases}$$

TABLE 7 Actual travel time and route type of routes between Houhai—Huaqiangbei.

Route no.	Route	The number of 'Left-behind'	Actual travel time/(s)	Route type
1	Line 11–Line 2	1.8	1650	The least time
2	Line 11–Line 3	1.2	1944	Less time
3	Line 2	1.1	2092	The fewest transfer
4	Line 11–Line 14– Line 7	2.5	2100	Less time
5	Line 11–Line 1	1.9	2208	Less time
6	Line 11–Line 7	0.6	2412	The least congestion

TABLE 8 Travel profiles of the target passenger.

Origin category	Destination category	Inbound period	Type of route	Route preferences
Corporate Enterprise	Corporate Enterprises	Weekday morning	The least time	0.29
Corporate Enterprises	Corporate Enterprises	Weekday morning	The fewest transfer	0.57
Corporate Enterprises	Corporate Enterprises	Weekday morning	The least congestion	0.14
…	…	…	…	…

TABLE 9 Comparison of route recommendation methods in the OD pairs of Houhai—Huaqiangbei.

Route recommendation category	Morning peak hours	Flat peak hours	Evening peak hours
Traditional Route Recommendations	Least time	Least time	Least time
Passengers' Real Route Choice	Fewest transfer	Least time	Least congestion
Personalized Route Recommendations	Fewest transfer	Least time	Fewest transfer

OD, origin-destination.

5.3 | Result analysis

Traditionally, route recommendations for passengers are mainly relied on generalized travel cost models, considering factors such as travel time and ticket fare. Typically, the route with the lowest cost is selected as the recommended route. However, personalized route recommendation methods have been developed to incorporate individual passenger preferences and congestion conditions during peak hours. Table 9 presents a comparison between the two conventional recommendation methods and the proposed method. The findings indicate that personalized route recommendation during peak hours is better equipped to satisfy the travel preferences of passengers, as illustrated in Figure 7.

The comparison during different time periods reveals that personalized route recommendation is more effective in meeting passengers' needs during peak hours. Conversely, the results of personalized and traditional route recommendation are the same during non-peak hours. This disparity can be attributed to the issue of passenger congestion during peak hours, which can be more accurately calculated and considered in personalized route recommendations. Traditional allocation models, such as the Logit model, lack the ability to calculate passenger route choices in congested situations, and therefore can only recommend the route with the lowest travel cost.

Subsequently, the personalized route recommendation is compared with the traditional recommendation methods for passengers at several OD pairs in Shenzhen Metro, as illustrated in Table 10. This evaluation allows for an assessment of the effectiveness and superiority of the personalized route recommendation approach over the traditional method for passengers at various stations.

6 | CONCLUSIONS

This paper proposes a personalized recommendation algorithm that incorporates passenger travel profiles and actual travel time with congestion. By utilizing cosine similarity, the algorithm identifies passengers with similar travel profiles. The case study demonstrates that during off-peak hours with low congestion, traditional and personalized route recommendations tend to yield similar results. However, during peak hours, personalized route recommendations prove more suitable for passengers' actual travel conditions. This method can be implemented in cities with significant congestion during peak hours and complex URT networks. Collaborating with map navigation software, URT operators can provide personalized route recommendations to passengers. By considering these factors, personalized route recommendation offers a more

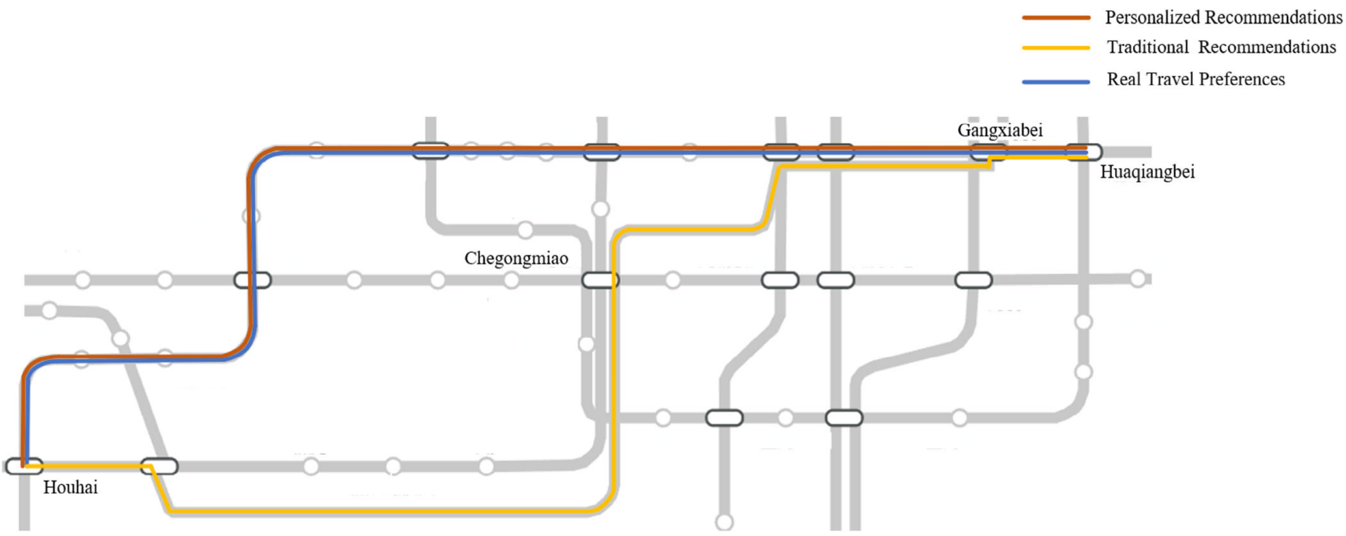

FIGURE 7 Comparison of traditional recommendations and the proposed method.

TABLE 10 Comparison of two route recommendations for several OD pairs.

OD	Traditional route recommendations	Personalized route recommendations	Passengers' real route choice
Shixia—Pingshanwei	Least time	Fewest transfer	Fewest transfer
Hongshan—Hongling	Least congestion	Fewest transfer	Fewest transfer
Shenzhenbei—Laojie	Least time	Least time	Fewest transfer
Bihaiwan—Futian	Fewest transfer	Fewest transfer	Fewest transfer
Bagualing—Hongshuwannan	Least time	Fewest transfer	Fewest transfer

OD, origin-destination.

sophisticated and individualized approach, aligning with passengers' travel preferences and improving the overall efficiency and user experience of the URT system.

However, the algorithm proposed in this paper does have limitations. For instance, the accuracy of the similar passenger judgment relies on a substantial amount of data. While this study utilized 1 month of data, it is recommended to use a year's worth of data for more precise calculations. Furthermore, the method only accounts for the similarity of passenger route choices and does not consider meticulous time granularity.

AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: **Wei Li**: Conceptualization; data curation; methodology; project administration; visualization; writing—review & editing. **Zhiyuan Li**: Data curation; formal analysis; validation; visualization; writing—original draft. **Qin Luo**: Funding acquisition; methodology; project administration; supervision; writing—review & editing. All authors reviewed the results and approved the final version of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The author elect to not share data.

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